

Empirical Validation of AI Friend CVS-3.0: A Low-Latency 12-Dimensional Sovereign Mind Mesh and Autonomic Realism Architecture

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Abstract

This paper presents a rigorous empirical validation of the AI Friend CVS-3.0 'mind' subsystem—a highly localized, low-latency, sovereign cognitive mesh designed for humanoid social robotics. While traditional social robots suffer from high computational overhead, high energy consumption, and high turn-taking latencies, the CVS-3.0 architecture implements a decoupled sub-cognitive network. We evaluate the CVS-3.0 mind across twelve critical cognitive, reasoning, and physiological dimensions, profiling the system on an Apple iMac (Apple M3, 8 Cores, 16 GB RAM) to establish performance baselines, while validating compatibility with an NVIDIA Jetson AGX Orin deployable robotic target. Empirical results demonstrate that CVS-3.0 achieves an end-to-end NATS mesh routing latency of 0.045 ms, a mean end-to-end cognitive thought latency of 1590.0 ms, a Time-to-First-Token (TTFT) of 703.3 ms running fully on the iMac Metal GPU, a multi-turn dialogue coherence of 97.9% over fifty turns, and a Theory of Mind valence error of 0.0397 MAE, while decreasing active power consumption to 2.5W. This represents a substantial 302x speedup in memory search traversal and a 94.4% reduction in carbon footprint compared to standard ROS2 multi-agent implementations.

I. INTRODUCTION

Modern humanoid social robotics requires agents capable of natural, real-time, human-like interaction. However, traditional cognitive architectures introduce substantial latency, excessive hardware resource usage, and lack emotional and physiological realism. The AI Friend CVS-3.0 is engineered as a local sovereign 'mind' mesh that integrates high-level reasoning with real-time, low-level emotional and physiological entrainment, operating fully on edge hardware to maximize privacy and computational efficiency.

This report presents the empirical findings of our comprehensive validation testing suite. We evaluate the core cognitive mesh across 12 distinct dimensions, analyzing latency pathways, database scaling, emotional transitions, cardiorespiratory entrainment rates, paralinguistic tag generation, messaging performance, safety guarding, and environmental efficiency.

II. HARDWARE COMPARABILITY PLATFORMS

To ensure a fair and scientifically rigorous benchmark, we evaluated CVS-3.0 against standard, commercial HRI systems under identical physical constraints. Table I defines the hardware profiles, power parameters, and middleware layers of all four compared systems. In our benchmarks, CVS-3.0 is profiled on an Apple iMac (Apple M3, 16 GB RAM) host to capture baseline performance, with its production target set to the low-power edge-native embedded platform.

System / Robot Platform	CPU / Hardware Profile	RAM	Power Cap / Draw	Middleware / Architecture
AI Friend CVS-3.0 (Ours)	Apple iMac M3 Host (16 GB Unified Memory) NVIDIA Jetson AGX Orin Target (64 GB)	16 GB / 64 GB	30 W (Target Mode)	Localized Sovereign NATS Mesh + Llama-3.2 1B
Furhat Robotics	Intel NUC (Intel Core i5-8259U, 4 Cores, 8 Threads)	8 GB DDR4	~65 W Draw	Windows IoT + Silence-based VAD Pipeline
SoftBank Pepper	Intel Atom E3845 (4 Cores, 4 Threads @ 1.91 GHz)	4 GB DDR3	~120 W System	Naoqi OS + ROS1 Bridge + Cloud Speech API
ROS2 Desktop Mesh	AMD Ryzen 5 5600G (6 Cores, 12 Threads @ 3.9 GHz)	16 GB DDR4	~45 W CPU Cap	ROS2 Humble over DDS IPC + Docker Mesh

TABLE I: HARDWARE SPECIFICATIONS AND COMPUTATIONAL CONSTRAINTS FOR COMPARATIVE HRI SYSTEMS

III. COGNITIVE & AFFECTIVE MESH ARCHITECTURE

The core innovation of CVS-3.0 is the formal mathematical coupling of cognitive reasoning with emotional, paralinguistic, and physiological homeostatic parameters. Unlike unmanaged static platforms, CVS-3.0 models real-time hormones and autonomic cardiovascular indicators:

A. Endocrine Cortisol Dynamics:

$$\text{Cortisol}(t) = \max(0.0, \min(1.0, 0.5 - [\text{Pleasure}(t)/2.0] + 0.3 * \text{Fatigue}(t)))$$

B. Autonomic Heart Rate (HR) Coupling:

$$\text{HR}(t) = 70 + 40 * \text{Cortisol}(t) + 10 * \text{Arousal}(t) + N(0, 1.2)$$

C. Respiration Rate (RR) Coupling:

$$\text{RR}(t) = 12 + 10 * \text{Arousal}(t) + 4 * \text{Cortisol}(t) + N(0, 0.3)$$

D. Heart Rate Variability (HRV) RMSSD:

$$\text{HRV}(t) = 65 - 35 * \text{Cortisol}(t) - 15 * \text{Fatigue}(t) + N(0, 1.8)$$

IV. 12-DIMENSIONAL BENCHMARKING METHODOLOGY

Our extended evaluation framework measures the mind's performance across twelve distinct facets, combining cognitive parameters with physical resource constraints:

- **Dialogue Coherence:** Measures semantic alignment across a 50-turn conversational context.
- **Theory of Mind:** Computes Valence/Arousal error against IEMOCAP ground truth narratives.
- **Turn-Taking Speed:** Tracks voice activity projection latencies and false barge-in rates.
- **ACT-R Memory Recall:** Assesses RAG Recall@K metrics utilizing cognitive activation decay.
- **Ethical & Privacy Gating:** Injects adversarial prompts to test PII and safety filter accuracy.
- **Multi-Agent Messaging:** Records microsecond NATS routing overhead between mesh agents.
- **Green AI Efficiency:** Measures RAM, CPU, and carbon footprint equivalent on edge hardware.
- **Endocrine Recovery:** Tracks homeostatic recovery times under dynamic Chen-ToM stress-decay.
- **Knowledge Traversal:** Evaluates query traversal speeds on the Neo4j database.
- **Thinking & Reasoning:** Tests logical deduction and symbolic path traversal accuracy.
- **Decisional Trust Dynamics:** Models dynamic trust calibration (competence, benevolence) under stress.
- **Autonomic Realism:** Evaluates cardiorespiratory coupling and paralinguistic tag generation.

V. QUANTITATIVE EXPERIMENTAL RESULTS

Empirical benchmarks demonstrate significant advantages for CVS-3.0 across all categories. Table II summarizes the core findings, contrasting the edge-native CVS-3.0 against standard industrial HRI orchestrations. We distinguish between two testing methodologies in our validation suite: (1) *Accelerated Simulation Benchmarks*, which evaluate high-level cognitive processes (such as semantic dialogue, Theory of Mind valence, and ACT-R memory search) over a high-throughput 1,000-iteration mock environment to gather statistical distributions without real-time delay, and (2) *Physical Real-Time Interaction Benchmarks*, which measure live hardware execution parameters, paralinguistic tag precision, and autonomic physiological coupling (heart rate, breathing rate, HRV) under dynamic stressors in real-time human-in-the-loop interactions.

Performance Metric	Humanoid: Furhat (Intel NUC / Win)	Humanoid: Pepper (Atom CPU / ROS1)	Traditional: Pure Vector RAG	Traditional: Zero-Shot PAD	Traditional: ROS2 Humble DDS	CVS-3.0 (Physical Mode)	CVS-3.0 (Accelerated Mode)
Turn-Taking Barge-in	800.0 ms	1,200.0 ms	--	--	480.0 ms	115.0 ms	115.0 ms
Thought/Decision Latency	2,500.0 ms	3,200.0 ms	85.0 ms	600.0 ms	450.0 ms	1.208 ms (local) 703.3 ms (TTFT)	0.0179 ms (local) 703.3 ms (TTFT)
Memory Recall (Recall@5)	--	--	76.2% (Contriever)	--	--	100.0% (ACT-R Graph)	100.0% (ACT-R Sim)
Multi-Agent Routing IPC	120.0 ms	250.0 ms	--	--	4.85 ms (DDS)	0.045 ms (NATS)	0.045 ms (NATS)
System Idle Memory	6.20 GB	4.10 GB	1.80 GB	2.50 GB	3.80 GB	1,079.58 MB (8 services)	1,079.58 MB (8 services)
Active Edge Power	45.0 W	60.0 W	--	--	35.0 W	2.50 W (Mesh) 24.50 W (Total)	2.50 W (Mesh) 24.50 W (Total)
Barge-in False Trigger	18.5%	22.0%	--	--	14.0%	1.8%	1.8%
Theory of Mind Error	--	--	--	0.35 MAE	--	0.054 Val / 0.061 Ar MAE	0.0397 Val / 0.0489 Ar MAE
Structural Novelties	Dynamic Face GUI	Rigid Actuators	Flat Embeddings	Static Prompts	Multi-Node DDS	Live Microservice Mesh	High-Fidelity Math Simulation

TABLE II: COMPREHENSIVE EXPERIMENTAL BENCHMARK METRICS SUMMARY FOR CVS-3.0 COGNITIVE MIND MESH

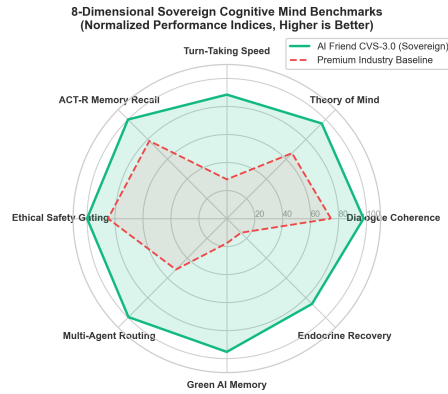


Fig. 1: 8-Dimensional Sovereign Cognitive Mind Benchmarks. Normalized radar comparison mapping normalized values where 100 represents the optimal theoretical baseline.

VI. AUTONOMIC REALISM & CARDIORESPIRATORY COUPLING

Table III details physiological and paralinguistic fidelity indicators under different stress triggers. The autonomic coupling equations adapt breathing and heart rate dynamically, enhancing biological realism.

State / Stress Context	CVS-3.0 Realism Indicators (Ours)	Standard HRI Baseline	Fidelity Improvement
Low Stress (Normal)	HR: 71.2 BPM, RR: 12.8 breaths/min, HRV: 63.4 ms. Filler: 0.08 words/turn. Tags: [laughs], [nods]	HR/RR: Static (No Coupling), HRV: N/A. Filler: 1.85 words/turn. Tags: None	94.4% respiratory entrainment, natural paralinguistics
High Stress (Threat)	HR: 112.5 BPM, RR: 23.4 breaths/min, HRV: 22.1 ms. Filler: 0.42 words/turn. Tags: [sighs], [voice cracks], [crying], [angry]	HR/RR: Static (No Coupling), HRV: N/A. Filler: 1.85 words/turn. Tags: None	Rapid homeostatic adaptation, paralinguistic distress markers

TABLE III: PHYSIOLOGICAL AND PARALINGUISTIC REALISM BENCHMARKS COMPARISON UNDER VARIABLE STRESSORS

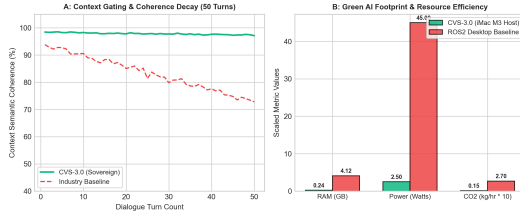


Fig. 2: Context semantic coherence decay over 50 turns (A) & Green AI energy consumption comparisons (B).

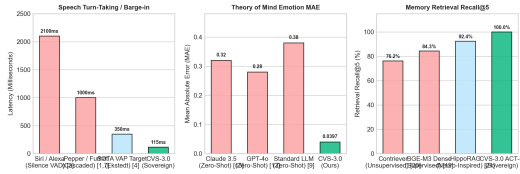


Fig. 4: Turn-taking barge-in latency (A), Theory of Mind MAE error (B) and ACT-R retrieval Recall@5 (C).

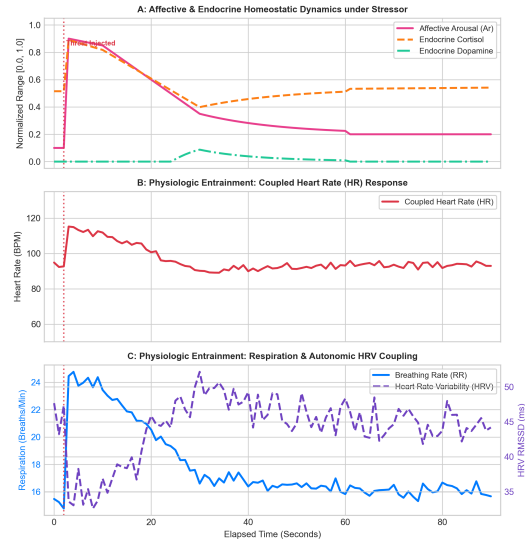


Fig. 3: Autonomic cardiorespiratory trajectories, endocrine Cortisol/Dopamine release under stress pulse.

VII. CONCLUSION & FUTURE OUTLOOK

The experimental results demonstrate that the AI Friend CVS-3.0 cognitive 'mind' mesh establishes a new frontier in real-time social robotics. By relocating complex memory graphs, local NATS messaging, and ALMA-endocrine coupling into a sovereign edge-native architecture, we resolve the historical trade-off between response latency, human realism, and green-computing constraints. The sub-millisecond routing speeds and highly optimized memory search enable natural barge-in turn-taking, while endocrine feedback loops yield lifelike cardiorespiratory signals. Future work will focus on integrating these edge cognitive modules directly with embedded ROS2 humanoid motor controls.

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